VANIER
CÉGEP/COLLEGE

Grocery Store Inventory Manager

January 26, 2026

Green Team
James Luciano
Daveena Patel
Mohamed Amine Hamidouch

Client Name: Reeba Shad
La Fruiterie Global
4625 Rue de Salaberry, Montréal, QC

Statement

We will be using code that has been developed in a previous course's (Application Development 1: Desktop) final project.

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Executive Overview

La Fruiterie Global is a grocery store located in Ahuntsic-Cartierville, Montreal. The client, Reeba Shad, seeks a system to manage her inventory, allowing her to easily track, view, manage products and details, as well as a user-friendly interface.

The client has ample experience with computer related matters as she is a software engineering student at Concordia University, which helps with a smooth communication/collaboration throughout the project. Our project currently aims to address difficulties related to inventory organisation.

The team will meet during scheduled class hours and use Github for control version as well as workflow management. Communication will overall take place on Instagram, with calls being made on Discord due to its superior call features.

The team operation will be guided by policies that emphasise respect, participation, open communication, and most importantly, adherence to the project plan. All responsibilities are fairly divided between the team members, and a different leader is selected to lead each deliverable. Deliverables include the project plan, requirements gathering and analysis, UML diagrams, prototype user interface, DB design, CI/CD pipeline, and implementation and client comments.

Overall, the project is organized to promote strong collaboration, defined responsibilities, and the successful delivery of a fully working and customized inventory management system.

Our Client

Description

Our client is Reeba Shad whose family owns a small local grocery store named La Fruiterie Global which is located in Ahuntsic-Cartierville, a borough of Montreal. They are looking for a system that can assist them in managing their inventory which will also provide them a clean and easy to manage way of viewing their products and all the details associated with them. Potential users using the proposed system include herself, her parents and any other employees who work at the store.

Computer Skills and Literacy

Client

The client has good computer skills. She is a software engineering student at Concordia University.

Parents of the Client

The client’s parents who work and manage the grocery store have very minimal Computer Skills and Literacy.

Employees

The employees of the client have minimal Computer Skills and Literacy.

Business Problem

As mentioned previously, the issue that our client was looking to solve is a system that allows them to manage the inventory of their establishment in an easy to manage, visually appealing and efficient manner. As a grocery store has many, many products and stock quantities, it would be beneficial if they had a system that would facilitate this process.

Organization

Description

Our team will be meeting up for project discussions, planning and working on Mondays from 8:30 to 11:30, Tuesday from 10:00 to 12:00 and Wednesdays from 3:30 to 5:30 as these are our class times for the course (System Development). If needed due to unforeseen circumstances we will discuss times available for out of class meetings. We will be using GitHub as an online repository in order to manage our workflow. For communication we will be primarily using Instagram for messaging, but we will do any video and/or audio calls on Discord as its call features are far superior to those of Instagram.

GitHub Link:
https://github.com/MohAmine2006/SystemDevProject.git

Team Policies

Policy	**Description**
We must follow our project plan	Throughout the duration of our project, each team member must respect the project plan and complete their tasks in the designated amount of time set. If needed, we can change the project plan to accommodate any unforeseen circumstances.
Participation is mandatory at ALL team meetings	Everyone must be present at each team meeting as they occur mostly during class time.
Everyone's opinions and ideas must be respected	For the sake of our team, we must ensure that every member feels heard and is allowed to express their ideas in order to discuss and have different perceptions of the task at hand.
Everyone must be in agreeance before we start a task/work	Similar to the task before, in order to keep the peace and be respectful, we must ensure that everyone is in agreement and is ok with any decision we decide to make. Whether it be the smallest or most significant task of the project.

Communication is key, if unsure about something ask for assistance	In order to prevent errors, if a member is having issues with any aspect or task throughout the duration of the project, they are encouraged to ask for help and discuss their troubles with the group.

Areas of Responsibility

Deliverables

*Note: The Team Leader is responsible for submitting reports of their assigned deliverable

Deliverable	**Team Leader**
Project Plan	Mohamed Amine Hamidouch
Requirements gathering and analysis	Daveena Patel
UML Diagrams	Daveena Patel
Prototype User Interface	James Luciano
DB Design	Mohamed Amine Hamidouch
CI/CD Pipeline	James Luciano
Implementation and client comments	Mohamed Amine Hamidouch

Areas of Application

Team Member	Area of Responsibility
James Luciano	Front-end
Daveena Patel	Back-end
Mohamed Amine Hamidouch	Database
Pavel Krivonos	N/A

Team Members Contact Information

Name	Email	Phone Number
James Luciano	jamesxluciano@gmail.com	438-526-6501
Daveena Patel	daveenampatel@gmail.com	514-572-6803
Mohamed Amine Hamidouch	m.a.hamidouch1@gmail.com	438-276-7757
Pavel Krivonos	N/A	N/A

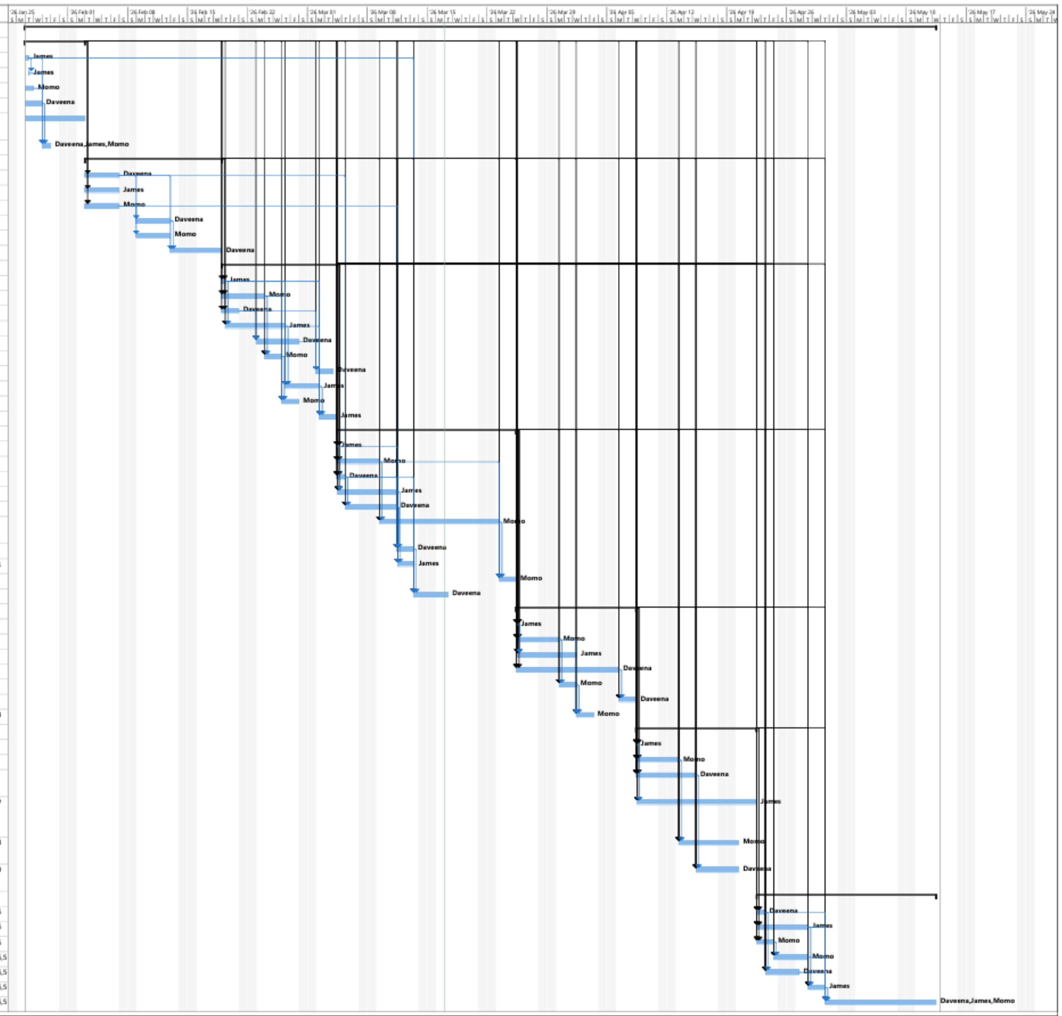
Client Contact Information

Client Name	Email	Phone Number
Reeba Shad	reeba1505@gmail.com	514-692-9219

Project Plan

Gantt Chart

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors
1			**Green Team**	**76.5 days**	**Tue 26-01-27**	**Wed 26-05-13**	
2			**Deliverable 1**	**5 days**	**Tue 26-01-27**	**Mon 26-02-02**	
3			Front matter	2 hrs	Tue 26-01-27	Tue 26-01-27	
4			Cover Page	2 hrs	Tue 26-01-27	Tue 26-01-27	3
5			Executive Overv	1 day	Tue 26-01-27	Tue 26-01-27	
6			Organization	2 days	Tue 26-01-27	Wed 26-01-28	
7			Refinement of details	5 days	Tue 26-01-27	Mon 26-02-02	
8			Project Plan,	1 day	Thu 26-01-29	Thu 26-01-29	3,4,5,6
9			**Deliverable 2**	**12 days**	**Tue 26-02-03**	**Wed 26-02-18**	
10			Basic questions	4 days	Tue 26-02-03	Fri 26-02-06	2
11			Business probler	4 days	Tue 26-02-03	Fri 26-02-06	2
12			questionare	4 days	Tue 26-02-03	Fri 26-02-06	2
13			Index 1	4 days	Mon 26-02-0	Thu 26-02-12	10
14			Index 2	4 days	Mon 26-02-0	Thu 26-02-12	12
15			Index 3	4 days	Fri 26-02-13	Wed 26-02-18	10,13
16			**Deliverable 3**	**9.25 days**	**Thu 26-02-19**	**Wed 26-03-04**	
17			Front matter	2 hrs	Thu 26-02-19	Thu 26-02-19	2,9
18			Executive Overv	3 days	Thu 26-02-19	Mon 26-02-23	2,9
19			Summary of Clie	2 days	Thu 26-02-19	Fri 26-02-20	2,9
20			Business probler	5 days	Thu 26-02-19	Thu 26-02-26	2,9,17
21			Narrarive	5 days	Mon 26-02-2	Fri 26-02-27	2,9,19
22			Appendix 1	2 days	Tue 26-02-24	Wed 26-02-25	2,9,18
23			Appendix 2	2 days	Mon 26-03-0	Tue 26-03-03	2,9,19,21
24			Appendix 3	2 days	Thu 26-02-26	Mon 26-03-02	2,9,17,20
25			Appendix 4	2 days	Thu 26-02-26	Fri 26-02-27	2,9,18,22
26			Appendix 5	2 days	Mon 26-03-0	Wed 26-03-04	2,9,17,20,24
27			**Deliverable 4**	**15 days**	**Wed 26-03-0**	**Wed 26-03-25**	
28			Front matter	2 hrs	Wed 26-03-0	Wed 26-03-04	2,9,16
29			Executive Overv	3 days	Wed 26-03-0	Mon 26-03-09	2,9,16
30			Revised summar	1 day	Wed 26-03-0	Thu 26-03-05	2,9,16
31			Business probler	5 days	Wed 26-03-0	Wed 26-03-11	2,9,16,28
32			usability guidelir	4 days	Thu 26-03-05	Wed 26-03-11	2,9,10,16,30
33			Copies of the prototype interfaces	10 days	Mon 26-03-09	Mon 26-03-23	2,9,16,29
34			Client's commer	2 days	Wed 26-03-1	Fri 26-03-13	2,9,16,30,32
35			Describtion the	2 days	Wed 26-03-1	Fri 26-03-13	2,9,12,16,28,31
36			Appendix 1	2 days	Mon 26-03-2	Wed 26-03-25	2,9,16,29,33
37			Appendix 2	2 days	Fri 26-03-13	Tue 26-03-17	2,3,9,16,30,34
38			**Deliverable 5**	**10 days**	**Wed 26-03-2**	**Wed 26-04-08**	
39			Front matter	2 hrs	Wed 26-03-2	Wed 26-03-25	2,9,16,27
40			Executive Overv	3 days	Wed 26-03-2	Mon 26-03-30	2,16,27,9
41			Business probler	5 days	Wed 26-03-2	Wed 26-04-01	2,9,16,27,39
42			Narrative descri	8 days	Wed 26-03-2	Mon 26-04-06	2,9,16,27
43			Appendix 1	2 days	Mon 26-03-3	Wed 26-04-01	2,9,16,27,40
44			Appendix 2	2 days	Mon 26-04-0	Wed 26-04-08	2,9,16,27,42
45			Appendix 3	2 days	Wed 26-04-0	Fri 26-04-03	2,9,16,27,40,43
46			**Deliverable 6**	**10.25 days**	**Wed 26-04-0**	**Wed 26-04-22**	
47			Front matter	2 hrs	Wed 26-04-0	Wed 26-04-08	2,9,16,27,38
48			Executive Overv	3 days	Wed 26-04-0	Mon 26-04-13	2,9,16,27,38
49			CI/CD Tool Selection and Justification	5 days	Wed 26-04-08	Wed 26-04-15	2,9,16,27,38
50			Pipeline Configuration and Design	10 days	Wed 26-04-08	Wed 26-04-22	2,9,16,27,38,47
51			Automated Testing Integration	5 days	Mon 26-04-13	Mon 26-04-20	2,9,16,27,38,48
52			Demonstration and Reflection	3 days	Wed 26-04-15	Mon 26-04-20	2,9,16,27,38,49
53			**Deliverable 7**	**15 days**	**Wed 26-04-2**	**Wed 26-05-13**	
54			Presentaion	1 day	Wed 26-04-2	Thu 26-04-23	2,9,16,27,38,46
55			Discussion	4 days	Wed 26-04-2	Tue 26-04-28	2,9,16,27,38,46
56			Future work	2 days	Wed 26-04-2	Fri 26-04-24	2,9,16,27,38,46
57			Appendix 1	2 days	Fri 26-04-24	Tue 26-04-28	2,9,16,27,38,46,5
58			Appendix 2	2 days	Thu 26-04-23	Mon 26-04-27	2,9,16,27,38,46,5
59			Appendix 3	2 days	Tue 26-04-28	Thu 26-04-30	2,9,16,27,38,46,5
60			Final revisions	9 days	Thu 26-04-30	Wed 26-05-13	2,9,16,27,38,46,5



Project: Project1
Date: Tue 26-03-17

Page 1

Agenda

First Meeting

Tuesday January 27 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
- We talked about how Pavel never showed up to class or tried to contact any members of the team
- We talked about the first deliverable

Decisions:
- We decided who the client will be for our project Daveena recommend a client and we decided to go with that
- We decided on who will do which part for each deliverable.
- We decided to use instagram and discord for communication
- We decided to create a github repository

Work Done:
- Created group chat on instagram and discord
- Created a github directory https://github.com/MohAmine2006/SystemDevProject.git
- We worked on the gantt chart as a team

Second Meeting

Wednesday January 28 2026
Location: D-241
3:30pm - 5:30pm

Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
- We talked to Alex about Pavel because he never contacted any of us
- We talked about the first deliverable

Decisions:
- We decided to work on deliverable 1

Work Done:
- We worked on deliverable 1

Third Meeting

Monday February 1 2026
Location: D-243
8:30-11:30

Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:

- We discussed how we'll start working on Deliverable 2 in the next coming days
- We discussed the feedback we got from Alex during our presentation

Decisions:

- We presented our project to Alex

Work Done:

- We presented Deliverable 1

Grocery Store Inventory Manager

February 16, 2026

Green Team
James Luciano
Daveena Patel
Mohamed Amine Hamidouch

Client Name: Reeba Shad
La Fruiterie Global
4625 Rue de Salaberry, Montréal, QC

Deliverable 2

Executive Overview

La Fruiterie Globale's business domain is the fruit and vegetable grocery store industry. Our client is Reeba Shad, she is the owner's daughter. Our client needs a better way to manage their inventory as they have a large extensive inventory with many different categories of items and as of right now their current way of monitoring and keeping track of their stock is extremely inefficient and tedious.

In this deliverable we set up a series of questions in order to gather the details and operations on how their business works and how we can assist them with our application. Some questions including how do they manage overstock and if how do they deal with expired products, that allow us to come up with the most effective and promising solution.

Daveena had a meeting with our client to ask questions about how they currently manage their inventory, the different types of categories of products they sell, what types of features they want and more questions related to how they currently do inventory.

Appendix 1, the user stories are different scenarios the employees and the owner will go through. For example the owner wants to be able to print the inventory information. The employees need to be able to update the inventory. The owner needs to be able to add, delete, and update the inventory information.

We represent said user stories in gherkin syntax in Appendix 2 this allows us to display in a simple format some scenarios where the desired features asked by our client will be useful on a day to day basis. These can include but are not limited to: logging in, updating product counts and removing a product from their inventory

Appendix 3, is the user stories mapping, this image clearly shows the user stories and what we need to work on for this project. It is an extraordinary layout that gives us insight on the different permissions and actions each user of this application has and will be able to use based on their needs and responsibilities.

Description

Business Domain

The business domain that we are engaging with is the fruit and vegetable grocery store industry which focuses on retailing fresh produce directly to the consumers via specialized stores or markets. This business sector is characterized by high-frequency, perishable inventory management which serves as the final step in the agricultural supply chain. This industry is heavily influenced by trends of healthy eating and local sourcing. Some other key aspects of this industry include market trends due to demand for organic products, supply chains as it involves importing/local sourcing to manage high perishability and competition with large supermarkets.

Client

The client is Reeba Shad, her family owns a store called La Fruiterie Global which is located in Ahuntsic-Cartierville, a borough of Montreal. She is a computer engineering student. This means that she is knowledgeable about IT but her parents and their employees are not that knowledgeable of IT. To manage their inventory they are currently using nothing, but just by looking to see what they are missing day by day and then proceeding to buy more of said product.

Business Problem

The business problem that our client is looking to solve is an improvement on managing the very large inventory of a smaller grocery store. Grocery stores, regardless of their size, sell plenty of items that make it very difficult in order to manage inventory and organize themselves to know which products they need to restock on. Having a visual web application that allows them to keep track of all their products with just a few clicks will greatly improve their efficiency when it comes to knowing their current product selection and stock amounts.

Questions

Questionaire

- How do you manage your inventory on a day-to-day basis?

  Answer: Their system of keeping track of inventory is multiple daily updates on paper, including overstock in the basement and updating records daily.

- How many different categories of food do you sell?

  Answer: Different categories of food they sell, including dairy, groceries, meat, frozen products, fruits, vegetables, cereals, jams, and other jarred items.

- How frequently do you update inventory records?

  Answer: It is done daily in the evening to prepare for the next day.

- What is the criteria that decides when an item needs to be reordered?
-

  Answer: Products that are replenished daily include fruits, while others like vegetables are replenished every two days.

- How are damaged or expired products handled?

  Answer: Their process for handling damaged products by significantly reducing their price and offering additional discounts if they can still be sold. For expired products, they check them before the expiry date and lower the price to ensure they are sold before they expire. They never sell expired products.

- How frequently do you replenish your inventory?

  Answer: Fruits and vegetables are replenished daily, while dairy products like milk and yogurt are replenished every two or three days.

- How often do you perform inventory counts in person?

  Answer: It is done daily by the workers and overseen by the managers.

- Who takes care of the inventory management, is it a manager or an employee?

  Answer: By the manager and not the employees.

- Do you want the inventory data to be available in a printable format?

  Answer: Yes they would like it to be.

- Are the managers/workers comfortable with using technology?
  Answer: They are comfortable using it.

- Do you ever run out of products unexpectedly?
  Answer: It happens occasionally, especially on busy days with high fruit sales.

- What languages would you like the system in?
  Answer: That both English and French are acceptable, but English is preferred.

- How would you like to view the lists of products on the application (ex: by category)?
  Answer: They would like them to be categorized by type, such as fruits and vegetables.

- Would you like alerts when stock is low?
  Answer: They would like alerts when stock is low.

- Do you have internet access in the store?
  Answer: Employees and managers have access, but not clients

- Do managers and employees have the same access?
  Answer: No employees can only update the inventory, while managers can add and delete products.

Open Questions

We didn't necessarily have any open questions so here is a brief recap:

On the inventory keeping front we know that they are using paper and roughly estimating their inventory numbers which is extremely inefficient and time consuming. However we do not currently know how their inventory counts are made and how frequently is being conducted each day and/or week. We assume that products which go bad more frequently such as fruits, vegetables and meats are prioritized, but we recognize that there is much more to running a grocery store than the product and meat sections. Many items are perishable, but some are not as well. It therefore becomes a crucial factor to determine which items are more frequently taken count of on a daily basis as well as know how often they go to restock these items during the week.

User Stories

Appendix 1

Employee user stories

Title: Update inventory
User Story: As an Employee I want to update the inventory information (price, count, expiration) in the evening so that we know how much inventory of a product is left.

Owner user stories

Title: Product Counts
User Story: As a store owner I want to be able increase/decrease the product count throughout the day so that I have an accurate product count at all times.

Title: CRUD Operations
User Story: As a store owner I want to be able add/delete a product so that I can manage what products our store still sells or not.

Title: Low Stock
User Story: As a store owner I want pop up notification alerts to know when to buy products before they are all sold out so that I don't lose customers.

Title: Over Stock
User Story: As a store owner I want pop up notification alerts to know when products are over stocked so I don't lose money and waste food.

Title: Profit
User Story: As a store owner I want to know how much of each product was sold in a day so that I know how much profit I made.

Title: Viewing Report
User Story: As a store owner I want a digital version of the inventory information so that I have a record of the information digitally.

Title: Printing Report
User Story: As a store owner I want a printable version of the inventory information so that I have a record of the information on paper.

User stories For both

Title: Log In
User Story: As a store owner or employee I want to be able to enter my username and password so I can log in to the application.

Title: Filter
User Story: As a store owner or employee I want to be able to filter the product display by category so I can view products from each category.

Title: Localization
User Story: As a store owner or employee I want to be able to switch the application language to English or French so I can correctly understand the information.

Appendix 2

Scenario: Successful update
Given I am in the app
When I login into the employee account
And I click on the update button
Then I should be able to update the inventory information (count, price) to have it up to date.

Scenario: Update inventory count when stock amount is changed
Given I am in the Owner homepage
When I access the inventory
And I change the count of a product
Then I should be able to update the count immediately.

Scenario: Adding new products to the list
Given I am in the Owner homepage
When I access the inventory
And I click on edit products
Then I should be able to add new products, or delete previous ones

Scenario: Products at risk of running out
Given I am in the Owner app homepage
When I enter the inventory page
And I click the alert button
Then I should be able to see all the products that I will run out from soon

Scenario: Products at risk of over stocking
Given I am in the Owner app homepage
When I enter the inventory page
And I click the alert button
Then I should be able to see all the products that I will soon be overstocked

Scenario: Ability to see total profit at the end of the day
Given I am in the Owner app homepage
When I enter the summary page
And I click on a date
Then I should be able to see the amount of products sold as well as the total profit

Scenario: Viewing a report
Given I am in the Owner homepage
When I click on the report button
And I select a date
Then I should be able to view the report

Scenario: Printing a report
Given I am in the Owner homepage
When I click on the report button
And I select a date
And I click the print button
Then I should be able to print a report for the day

Scenario: User Login
Given I am on the login page
When I enter my username and password
And I click the login button
Then I should be taken to the inventory homepage

Scenario: Viewing Products
Given I am on the inventory homepage
When I choose a product category to view (ex: fruits)
And I click on the button
Then I should be taken to the inventory homepage for that category

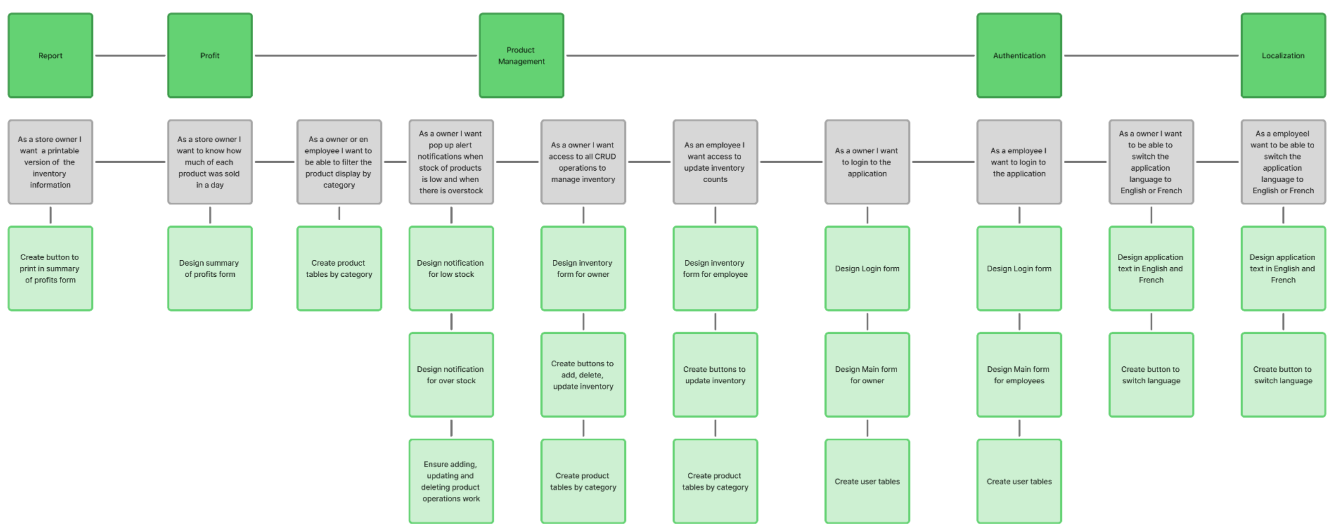
Scenario: Switching Language
Given I am on either the Employee homepage or the Owner homepage
When I click the language icon
And I select the language
Then I should view the application in the selected language (English or French)

Appendix 3

Explanation

We chose to use the design platform FigJam/Figma for our user story map as it is a simple to use application and also one of the first results that came up when we searched up for tools to create a user story map. It also has an excellent system of allowing us to work on it simultaneously. Figma is also a renowned collaborative design tool used by teams in order to build realistic diagrams which we used for our application user story map and its features.

Printout



Agenda

Fourth Meeting

Tuesday February 3, 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
We discussed deliverable 2
We discussed how we will meet the client

Decisions:
We decided to work on deliverable 2
We decided to do a zoom meeting with the client

Work Done:
Worked on deliverable 2

Fifth Meeting

Wednesday February 4, 2026
Location: D-241
3:30PM - 5:30PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
We discussed using Jira
When we will meet up with client

Decisions:
We decided only Daveena will meet up with client as she is not comfortable with a zoom meeting
We will be recording the meeting though so we know all answers

Work Done:
Worked on deliverable 2
We set up Jira/Confluence

Sixth Meeting

Tuesday February 10, 2026
Location: D-244
10:00 - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
We discussed the details of our user stories and which features to add
Alex discussed with us how Pavel has dropped the course
We discussed how we need a bit more details for the user stories/questionaire

Decisions:
Pavel Krivonos has dropped the course, he is gone

Work Done:
Continued working on deliverable 2
Got more details from the client
Got feedback from Alex on our user stories/questionaire
Finished our user stories, gherkin syntax and map
Started the executive overview

Seventh Meeting

Wednesday February 11, 2026
Location: D-241
3:30PM - 5:30PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Finishing up deliverable 2 for next week

Decisions:
Asked Alex for feedback on our user stories ok
We decided to use FigJam for our user story map ok
Gerkhin Syntax is ok

Work Done:
Continued working on deliverable 2

Eighth Meeting

Monday February 16, 2026
Location: D-243
8:30 - 11:30
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Feedback from Alex to finish the deliverable

Decisions:
Update user stories
Update gherkin syntax
Update user story map
Update Project Plan with appendices 1,2,3
Put User stories on jira

Work Done:
Updated deliverable 2
James updated gherkin syntax, user story map, user stories
Daveena update project plan
We all added user stories to jira plus descriptions

Grocery Store Inventory Manager

February 17, 2026

Green Team
James Luciano
Daveena Patel
Mohamed Amine Hamidouch

Client Name: Reeba Shad
La Fruiterie Global
4625 Rue de Salaberry, Montréal, QC

Deliverable 3

Executive Overview

La Fruiterie Globale's business domain is the fruit and vegetable grocery store industry. Our client is Reeba Shad, she is the owner's daughter. Our client needs a better way to manage their inventory as they have a large extensive inventory with many different categories of items and as of right now their current way of monitoring and keeping track of their stock is extremely inefficient and tedious.

For this deliverable we worked on different diagrams that will help us complete the project in order to satisfy the clients needs.

Appendix 1 is the flowchart system diagram which deals with flow of the application and how the users will navigate throughout the system.

Appendix 2 is the Use Cases UML diagram which shows the different uses between the types of users using this system.

Appendix 3 is the Sequence Diagram UML diagram which goes through each sequence throughout the application and its workflow.

Appendix 4 is the UML Class diagram which defines our classes, their properties and methods necessary for the implementation of the system..

Description

Client

The client is Reeba Shad, her family owns a store called La Fruiterie Global which is located in Ahuntsic-Cartierville, a borough of Montreal. She is a computer engineering student. This means that she is knowledgeable about IT but her parents and their employees are not that knowledgeable of IT. To manage their inventory they are currently using nothing, but just by looking to see what they are missing day by day and then proceeding to buy more of said product.

Business Problem

The business problem that our client is looking to solve is an improvement on managing the very large inventory of a smaller grocery store. Grocery stores, regardless of their size, sell plenty of items that make it very difficult in order to manage inventory and organize themselves to know which products they need to restock on. Having a visual web application that allows them to keep track of all their products with just a few clicks will greatly improve their efficiency when it comes to knowing their current product selection and stock amounts.

Narrative Description

The inventory is checked by the employee. The employee keeps a list of what needs to be ordered in the evening. They update the list during the day. They then tell the owner what needs to be ordered in the evening.

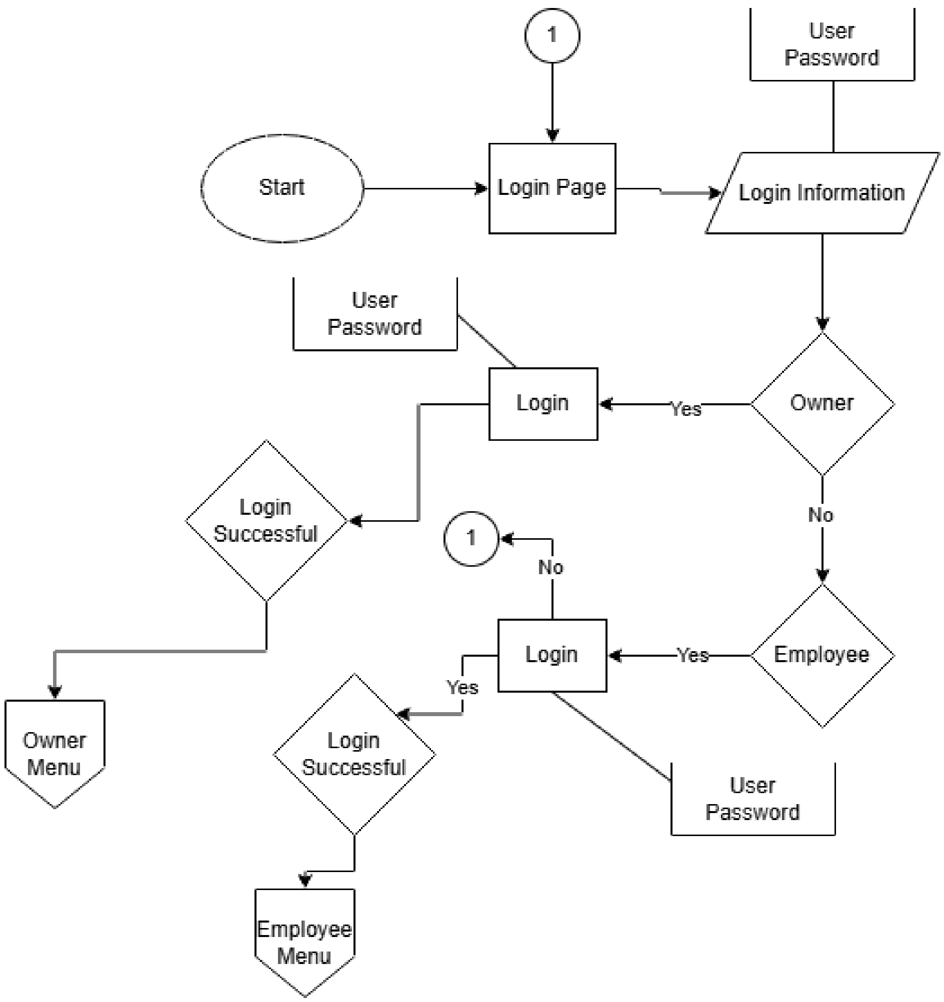
For example the employee tells the owner that milk was one of the products that was sold the most. Then the owner will check the inventory of milk and will decide if he needs to place an order. If the owner decided to place an order. He will call their supplier to place an order to have the products delivered. Then their supplier will deliver it to them. Once the product is delivered to the store the employee checks if the quantity is right and checks the expiry date. If the products are expired the owner contacts the supplier and the products are returned and replaced.If the supplier doesn’t deliver products on time the owner calls the supplier for updates, but nothing else can be done. Our client told us that they can’t do much.

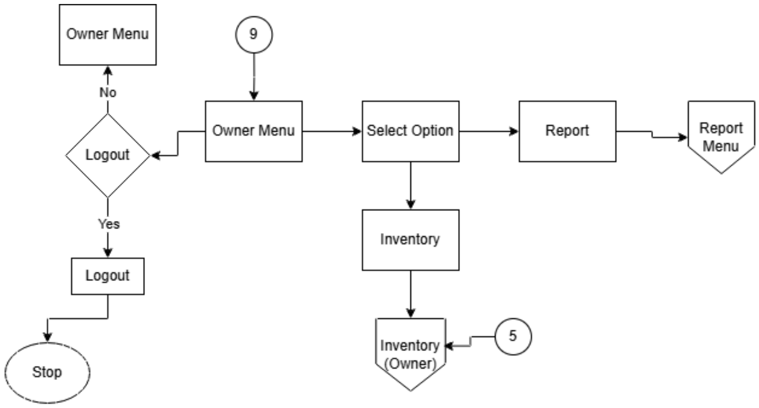
The employee is also responsible to inform the owner if a product is being sold more. In order for the owner to decide if he will order more of that product the next time he orders the product. It is also the same if a product is not as popular as it used to be. The employee is responsible to notice and tell the owner. Then the owner decided to order less of that product.

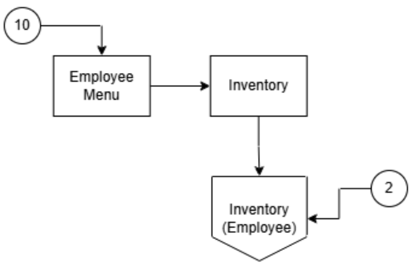
Diagrams

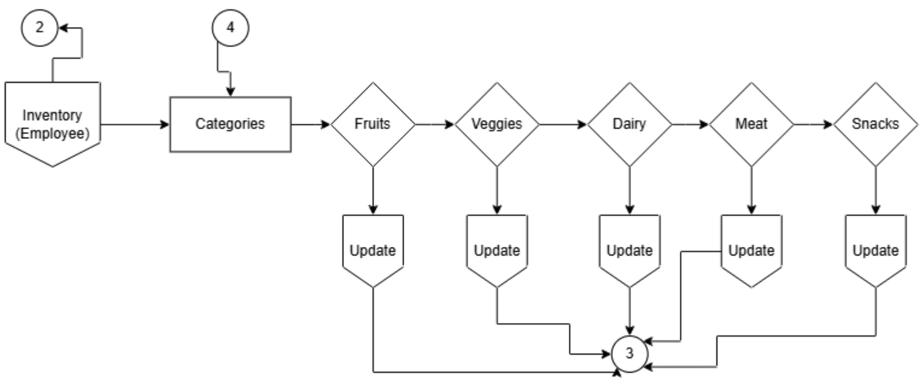
Appendix 1

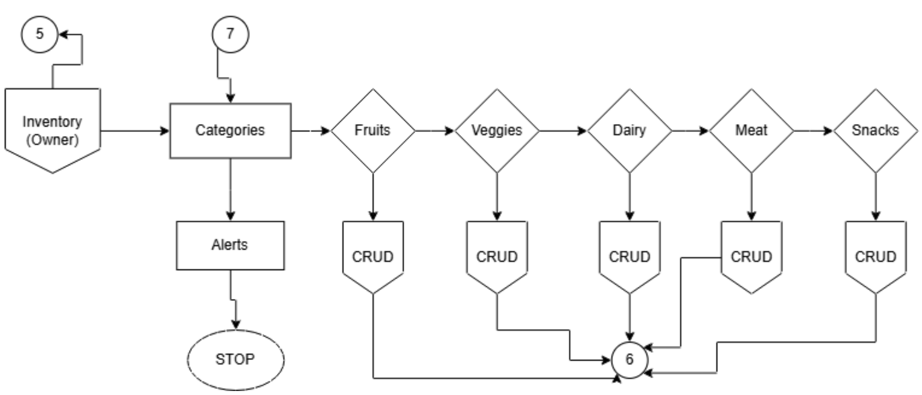
Flowchart System Diagram

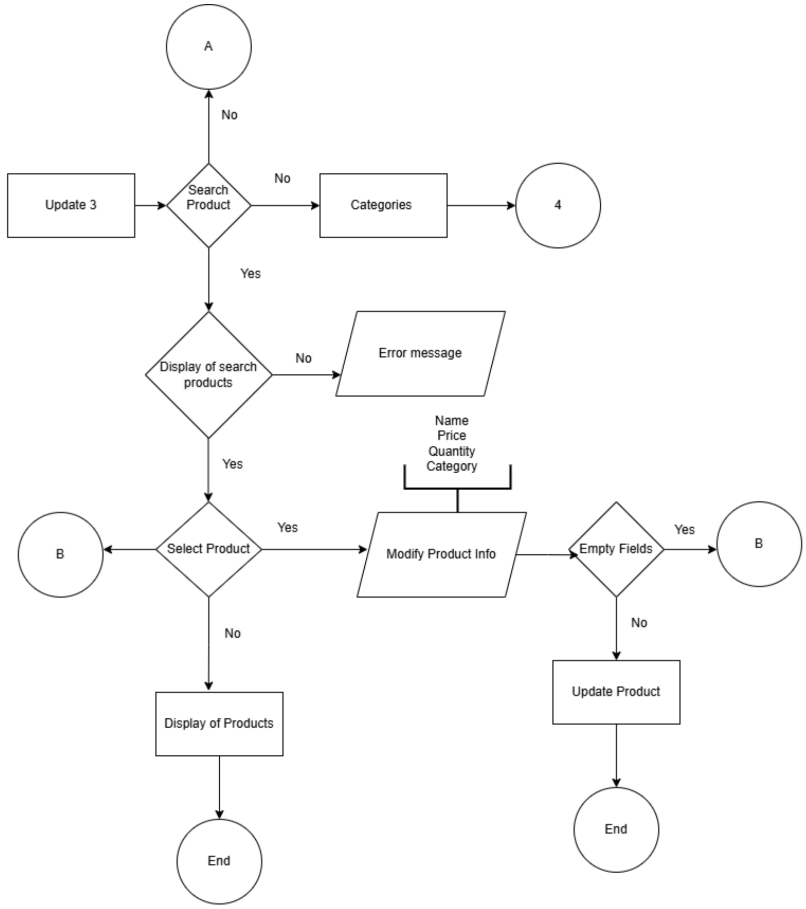


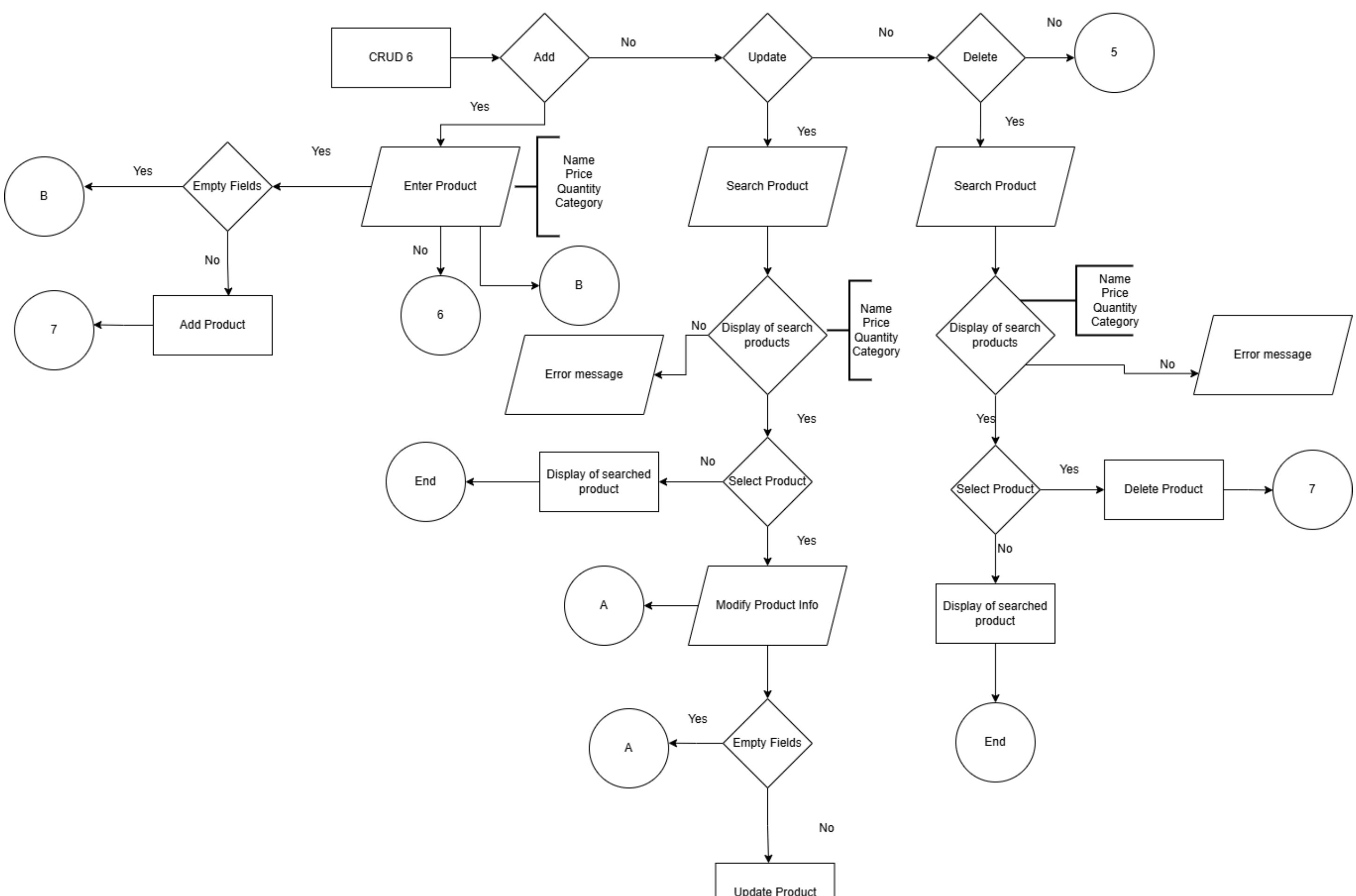

Owner Menu
9
No
Logout
Owner Menu
Select Option
Report
Report Menu
Yes
Inventory
Logout
Inventory (Owner)
5
Stop

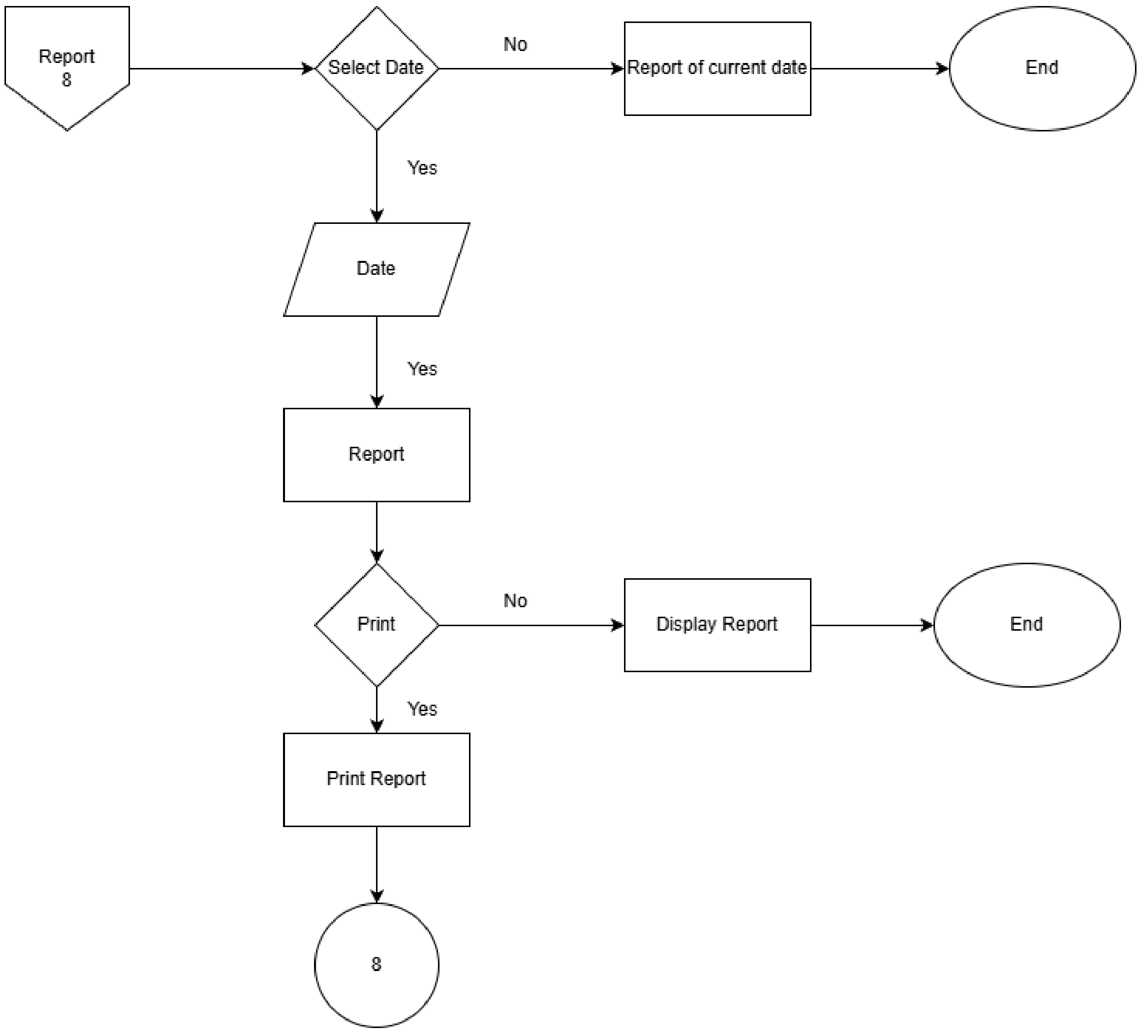

10
Employee Menu
Inventory
Inventory (Employee)
2


2
4
Inventory (Employee)
Categories
Fruits
Veggies
Dairy
Meat
Snacks
Update
Update
Update
Update
Update
3


5
7
Inventory (Owner)
Categories
Fruits
Veggies
Dairy
Meat
Snacks
Alerts
CRUD
CRUD
CRUD
CRUD
CRUD
STOP
6

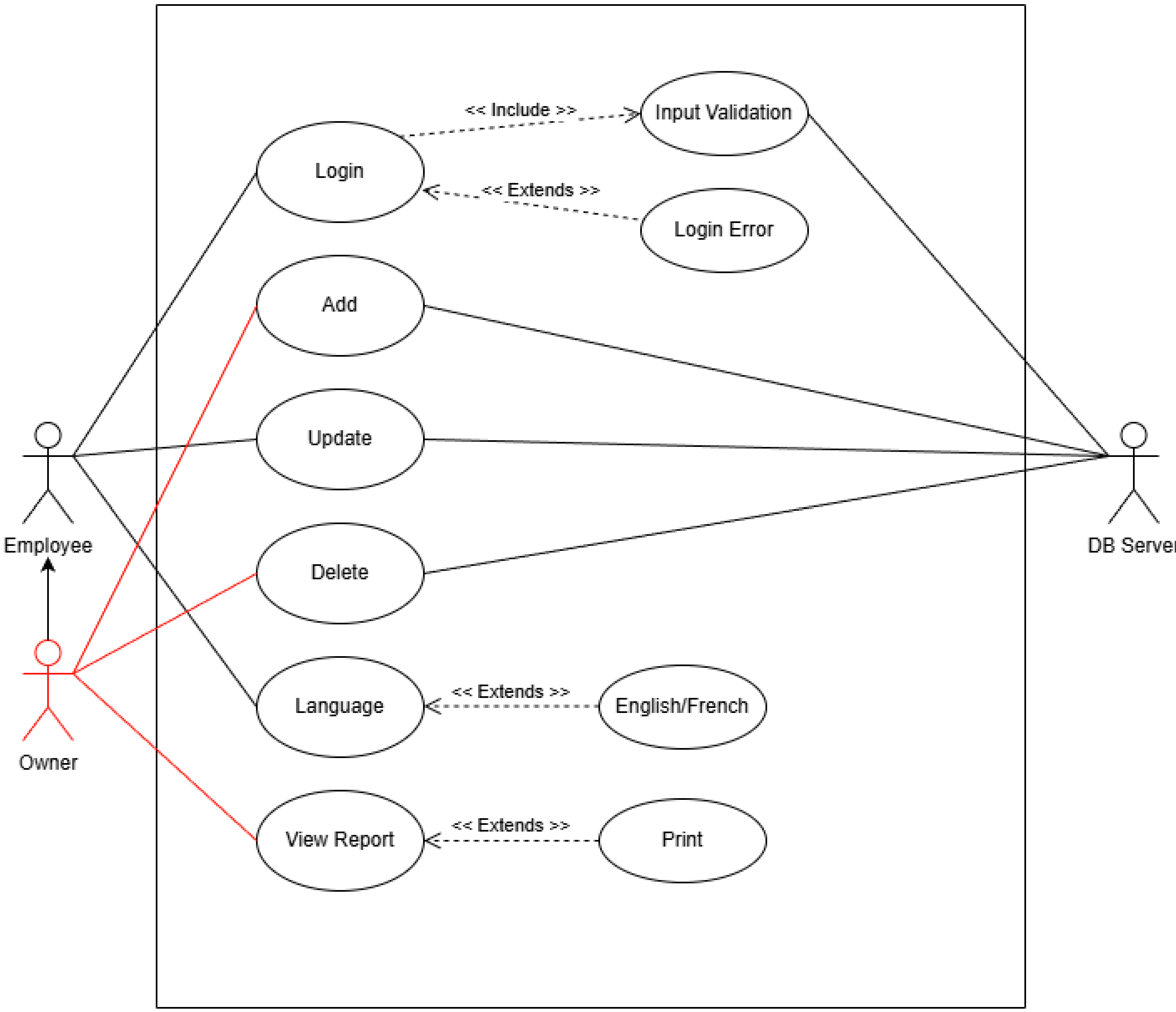

A
No
Update 3
Search Product
No
Categories
4
Yes
Display of search products
No
Error message
Yes
Name
Price
Quantity
Category
B
Select Product
Yes
Modify Product Info
Empty Fields
Yes
B
No
No
Display of Products
Update Product
End
End


CRUD 6
Add
No
Update
No
Delete
No
5
Yes
Yes
Yes
Yes
Yes
B
Empty Fields
Enter Product
Name
Price
Quantity
Category
Search Product
Search Product
No
No
B
Name
Price
Quantity
Category
Name
Price
Quantity
Category
7
Add Product
6
No
Display of search products
Display of search products
Error message
No
Error message
Yes
Yes
End
Display of searched product
No
Select Product
Select Product
Yes
Delete Product
7
Yes
No
A
Modify Product Info
Display of searched product
End
Yes
A
Empty Fields
No
Update Product


Report
8
Select Date
No
Report of current date
End
Yes
Date
Yes
Report
Print
No
Display Report
End
Yes
Print Report
8

Appendix 2

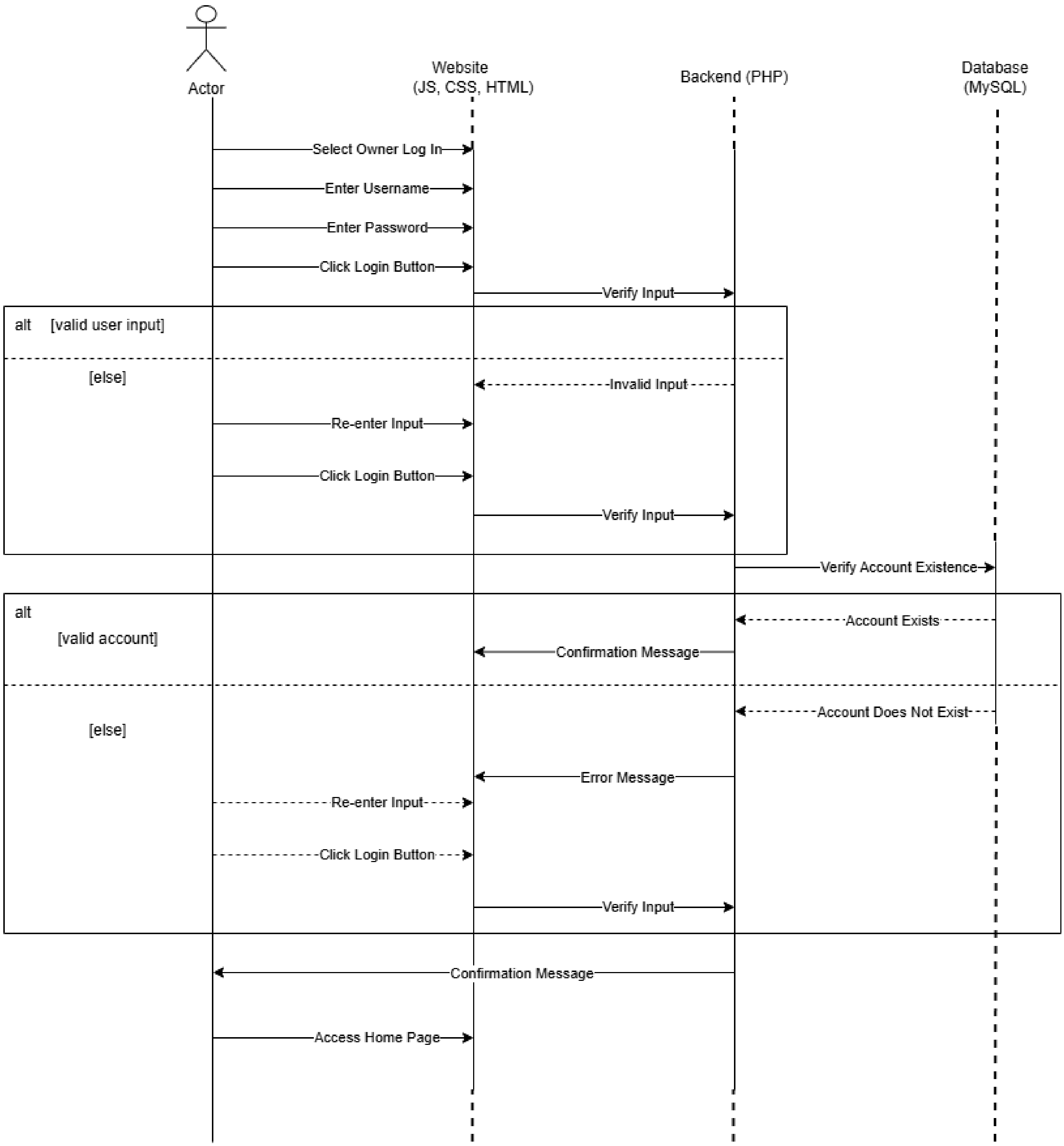
Use Cases UML Diagram



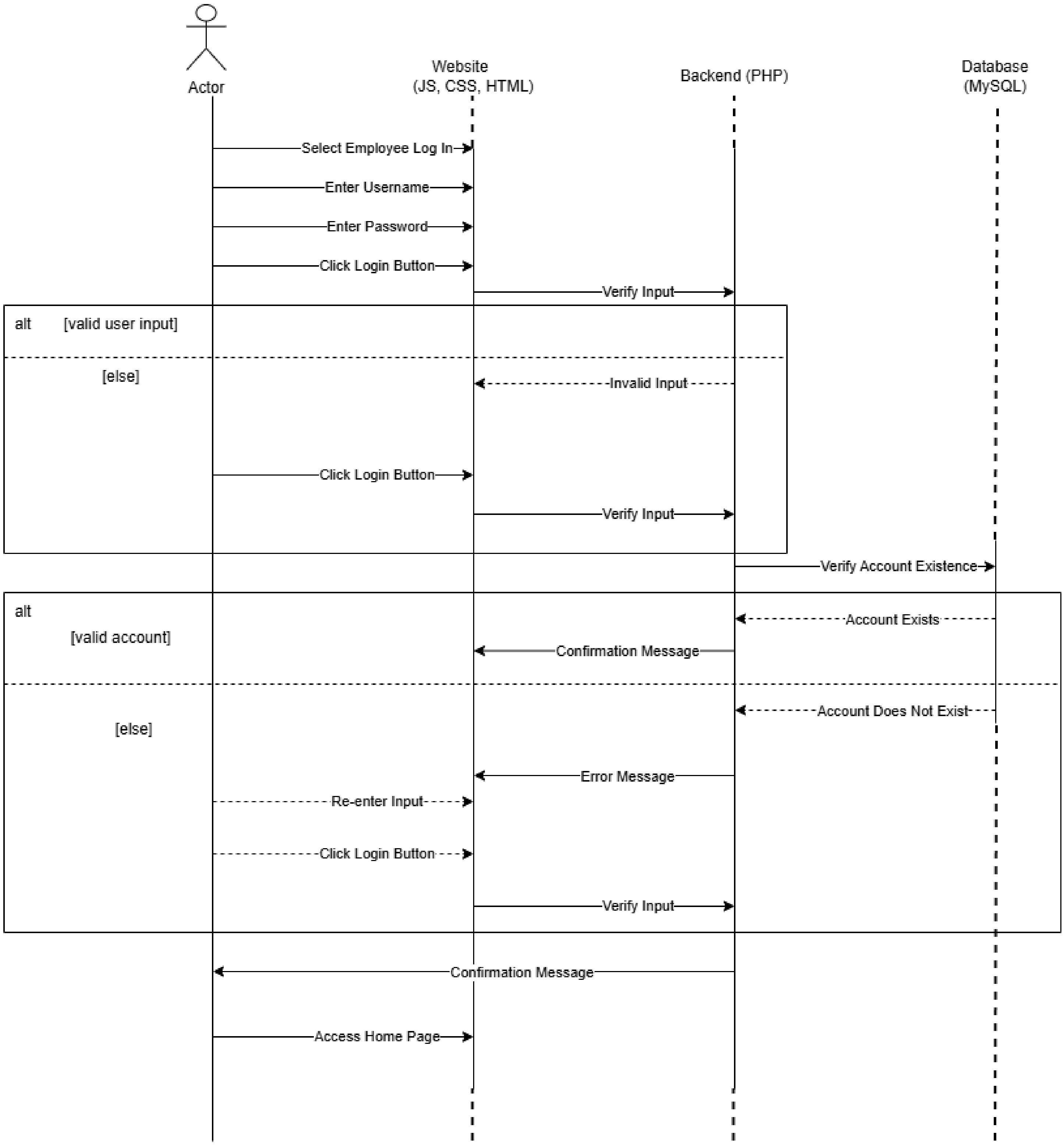
Appendix 3

Sequence Diagram UML Diagram

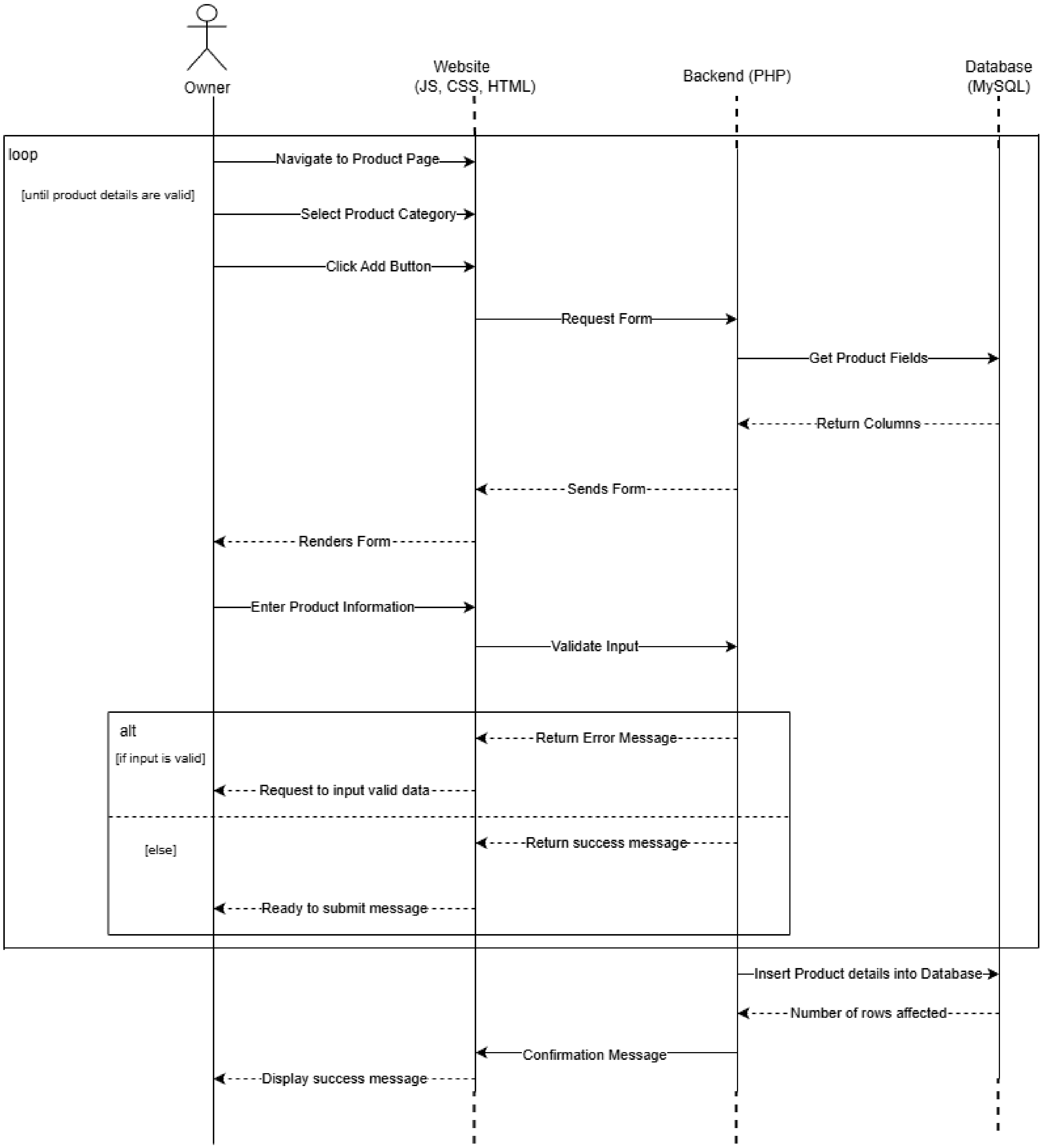
Owner Log in



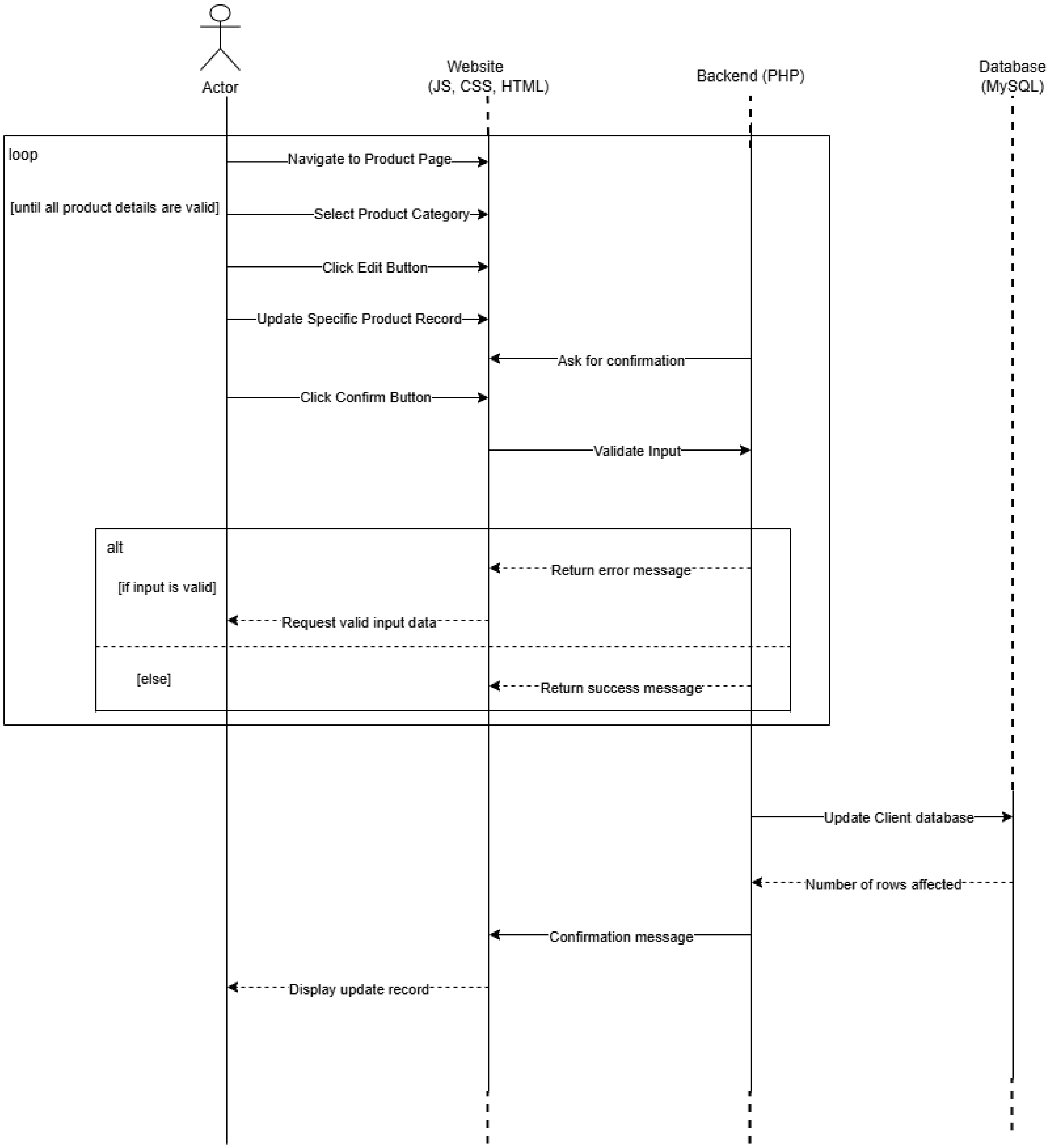
Employee Log in



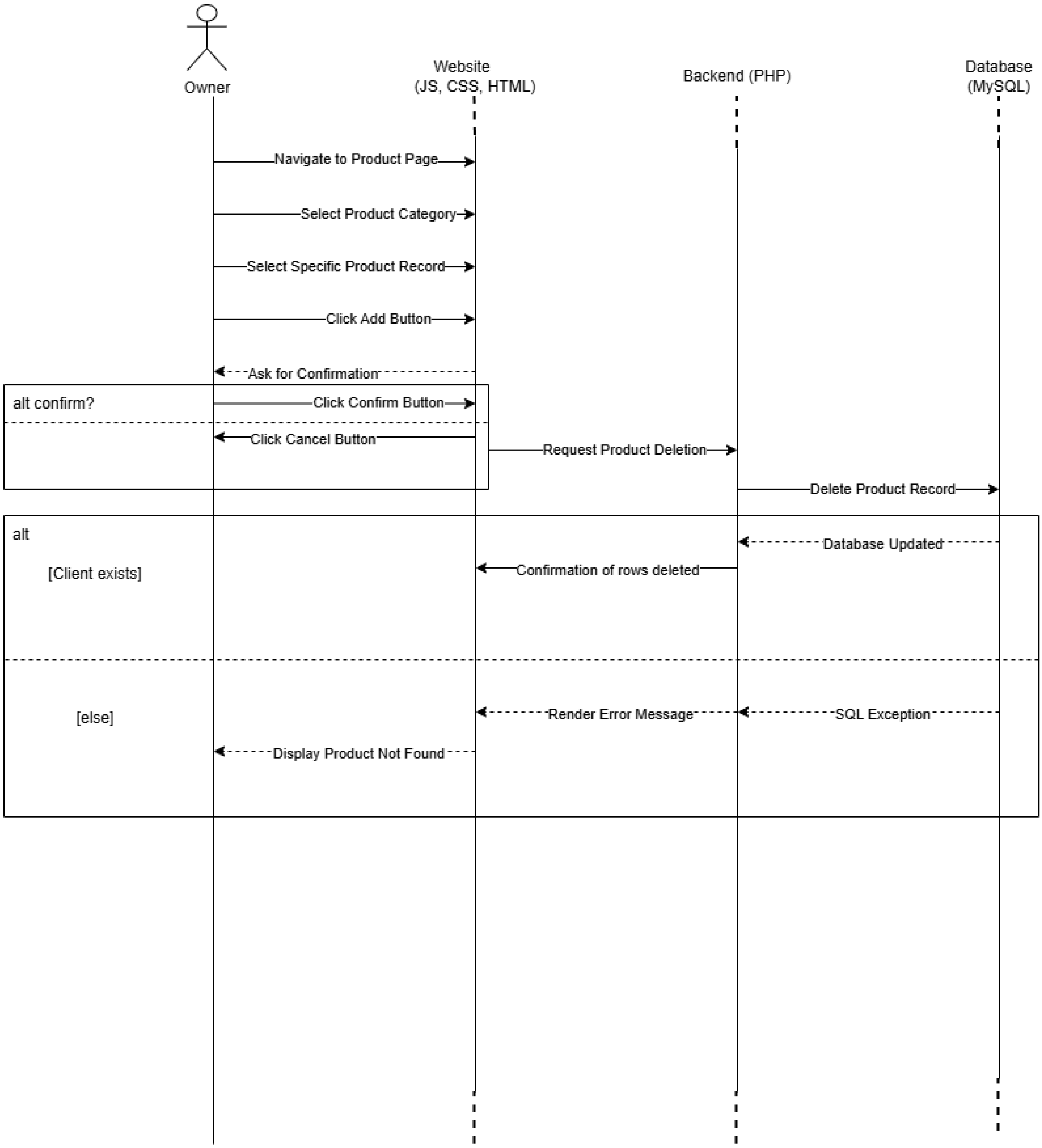
Add Product


Owner
Website
(JS, CSS, HTML)
Backend (PHP)
Database
(MySQL)
loop
[until product details are valid]
Navigate to Product Page
Select Product Category
Click Add Button
Request Form
Get Product Fields
Return Columns
Sends Form
Renders Form
Enter Product Information
Validate Input
alt
[if input is valid]
Return Error Message
Request to input valid data
[else]
Return success message
Ready to submit message
Insert Product details into Database
Number of rows affected
Confirmation Message
Display success message

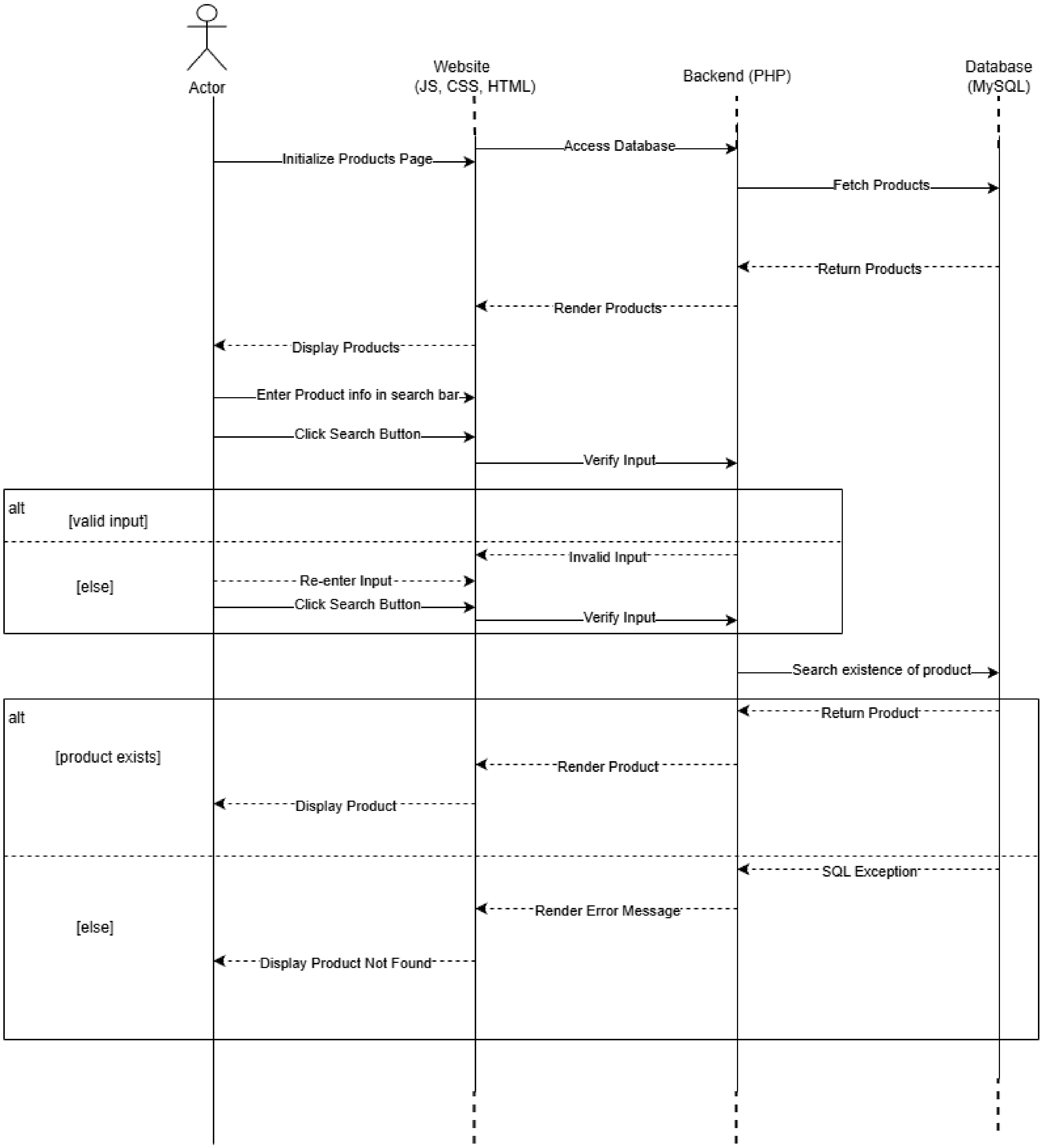
Update Product



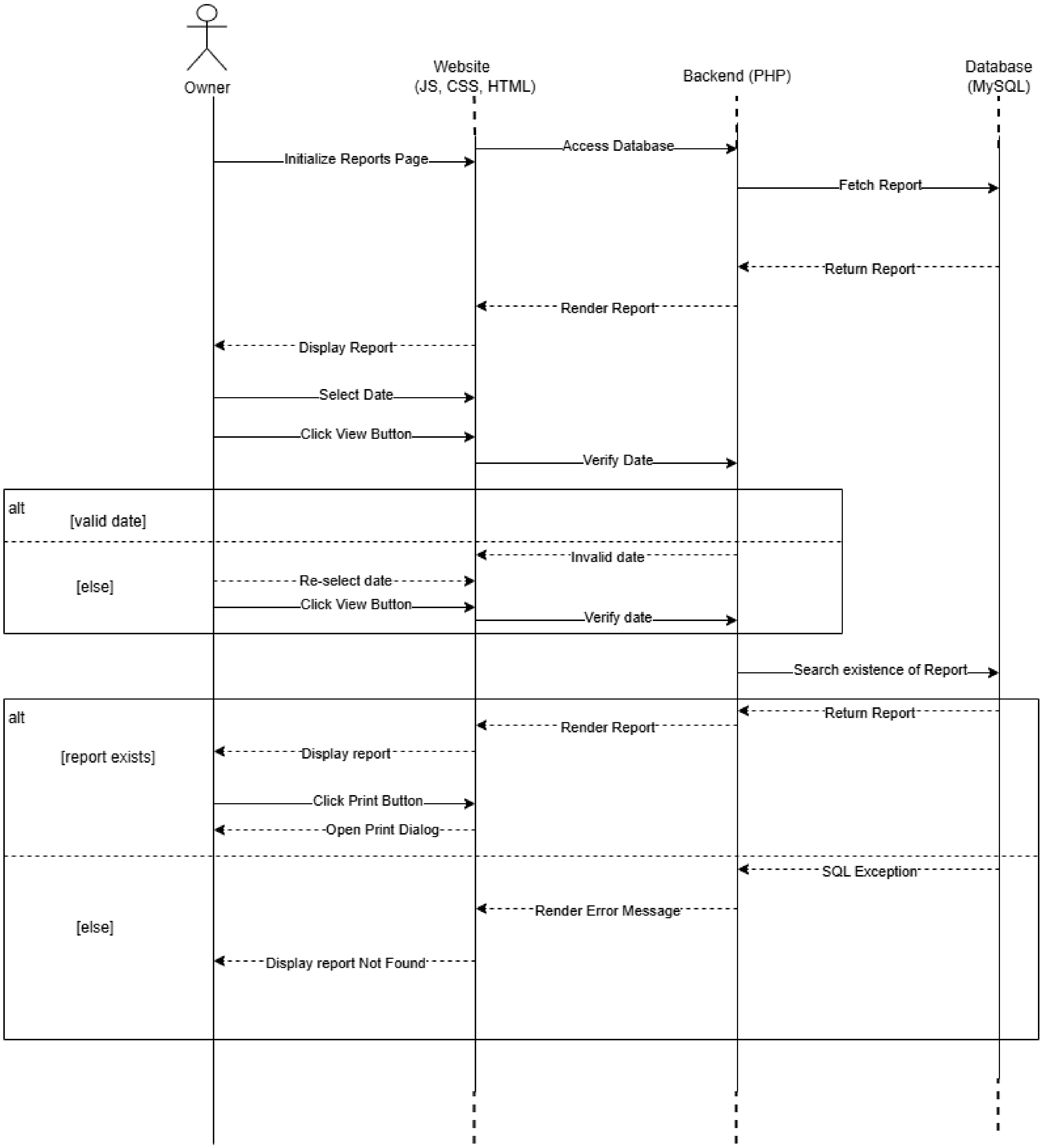
Delete Product



Search Product

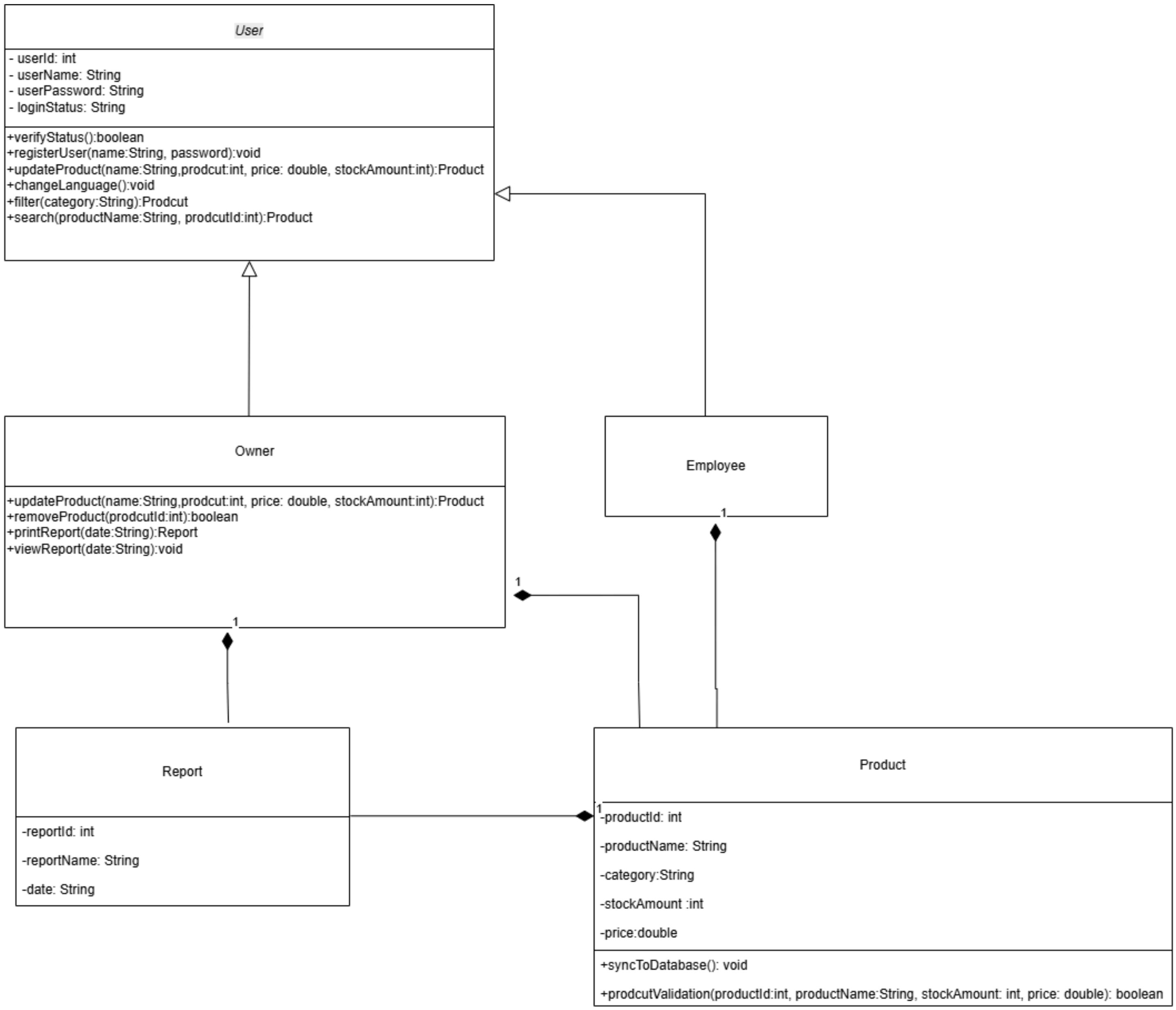


View Report


Owner
Website
(JS, CSS, HTML)
Backend (PHP)
Database
(MySQL)
Initialize Reports Page
Access Database
Fetch Report
Return Report
Render Report
Display Report
Select Date
Click View Button
Verify Date
alt
[valid date]
[else]
Invalid date
Re-select date
Click View Button
Verify date
Search existence of Report
alt
[report exists]
Return Report
Render Report
Display report
Click Print Button
Open Print Dialog
[else]
SQL Exception
Render Error Message
Display report Not Found

Appendix 4

UML Class Diagram



Appendix 5

Our client wasn’t able to provide copies of forms and documents they use, but she did say that they usually just write information on paper.

Agenda

Ninth Meeting

Tuesday February 17, 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Discuss how we will be doing the flowchart

Decisions:
Decided who will do which diagrams
Decided how we will do our flowchart

Work Done:
Worked on deliverable 3
Made a mock up flowchart to practice

Tenth Meeting

Wednesday February 18, 2026
Location: D-241
3:30PM - 5:30PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Continuing with flowchart
Discussed workplan for deliverable 3
Discussed future charts to be learned for deliverable

Decisions:
Asked Alex for feedback on flowcharts
Decided to ask client more details for narrative description
Decided to use draw.io for flowchart

Work Done:
Worked on deliverable 3
Continued making a mock up flowchart to practice
Started making flowchart online

Eleventh Meeting

Monday February 23, 2026
Location: D-243
8:30 - 11:30
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Our finished flow charts and how we'll put them in document
Use cases diagrams

Decisions:
Started use case diagrams and divided them among us

Work Done:
Worked on deliverable 3
Started use cases diagram

Twelfth Meeting

Tuesday February 24, 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Discuss how we will be doing the use case chart

Decisions:
Decided to continue working on use case diagram
Make mock up on paper
Asked Alex for feedback

Work Done:
Worked on deliverable 3
Finished mock use case diagram

Thirteenth Meeting

Monday March 2nd, 2026
Location: D-243
8:30 - 11:30
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Discussed how we will be making our UML class diagram (drew on paper)

Decisions:
Started UML class diagram
Asked Alex for feedback on it

Work Done:
Worked on deliverable 3
Started UML class diagram

Fourteenth Meeting

Tuesday March 3rd, 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Discuss how we will be doing the sequence diagram

Decisions:
Decided how to do sequence diagram

Work Done:
Worked on deliverable 3
Started sequence diagram
Finished class diagram

Fifteenth Meeting

Tuesday March 17, 2026
Location: D-244
10AM - 12:00PM
Participants: James Luciano, Daveena Patel, Mohamed Amine Hamidouch

Topics to Discuss:
Discuss finishing up feedback for deliverable 3

Decisions:
Corrected mistakes in flow chart and sequence diagrams

Work Done:
Finished corrections in deliverable 3